

Relation-aware heterogeneous graph neural network for entity alignment

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ABSTRACT

Entity alignment refers to finding equivalent entities from different knowledge graphs. Most of the existing entity alignment methods are studied based on homogeneous graphs. However, the knowledge graph is a heterogeneous graph containing many types of nodes, such as entities, relations, and attributes. Therefore, we propose introducing a heterogeneous graph neural network to model entities and relations simultaneously and propose an iterative fusion method to enhance the interaction between these two semantic nodes. Since not all datasets contain relation information, this paper does not directly introduce a feature representation of relations. The generalizability of the approach is improved by utilizing a relation-aware strategy to obtain information about the relation. Specifically, the information propagation of the head and tail entities in the triplet is utilized to obtain the feature representation of the relation. Experimental results show that the present method performs better on three cross-lingual datasets *DBP15K* and two large-scale datasets *DWY100K*.

1. Introduction

In recent years, knowledge graph (KG) has become a very hot research direction, and research results related to knowledge graph have been launched continuously [1–3]. Entity alignment, as an essential step in knowledge graph construction and fusion, aims to determine whether two entities from different knowledge graphs are equivalent. More specifically, it is to judge whether these two entities point to the same object in the real world. The accuracy and completeness of the entity alignment results will directly affect the final constructed knowledge graph's quality and indirectly determine the downstream tasks' performance. This also indicates that the link of entity alignment is crucial. Entities can be aligned mainly based on the assumption that entities adjacent to an aligned entity may also be aligned. As shown in Fig. 1, new aligned entities can be inferred in the knowledge graph if a portion of the aligned entities is known in advance.

There are two main types of traditional entity alignment methods: one is based on probabilistic models [4–6], and the other is based on machine learning [7,8]. However, with the increasing size of knowledge graphs, traditional entity alignment methods can no longer effectively align entities in different knowledge graphs and are thus gradually no longer used. As representation learning techniques begin to be widely used in knowledge graphs, some researchers have begun to apply representation learning to the entity alignment task and have performed well. Entity alignment methods using representation learning to obtain entity embeddings [9] have become mainstream.

Currently, embedding-based models are broadly classified into two categories: one is based on TransE [10–13]. This approach interprets the relations in a relation triplet as translation vectors from head entity to tail entity, and the low-dimensional embedding vector space reflects the semantic information of the entities. Another category is GNN-based models [14,15]. This approach treats the knowledge graph entities as nodes and relations as edges. It uses the edges to propagate information to obtain more expressive embeddings by fusing the entity's neighborhood information into the entity embeddings. One problem with this approach is that relations are used only as edges, ignoring the rich information in them. This may lead to poor performance on entity alignment with similar structures.

The two entities in Fig. 2 have the same structure, and the two central entities, “China” and “Beijing”, are aligned based on the structure information alone. However, in reality, the two entities are entirely different, which is the disadvantage of relying solely on structural information. In order to solve this problem, this paper adds relation information to the entity alignment task, and the entity alignment after considering the relation is shown in Fig. 3. After considering the relation information, although the two entities have the same surrounding structure, they do not have the same relation with the same neighboring entities. Therefore, it can be judged that the probability of these two entities not matching is relatively high, thus successfully avoiding mismatching.

This paper proposes an entity alignment method based on relation-aware heterogeneous graph neural networks to address the above

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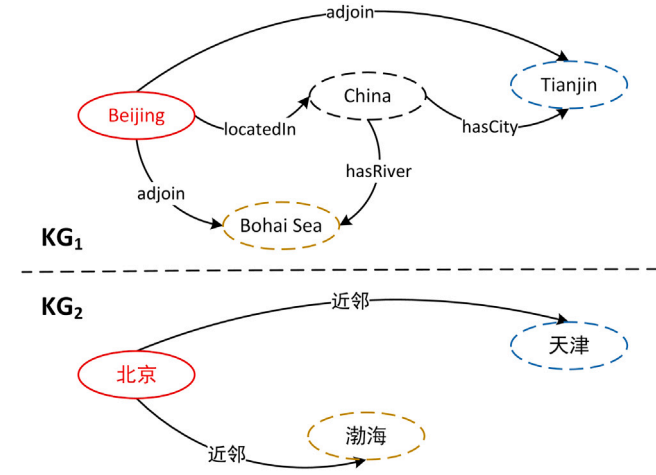


Fig. 1. Example of alignment of two heterogeneous KGs (Circles of the same color indicate priority alignment.).

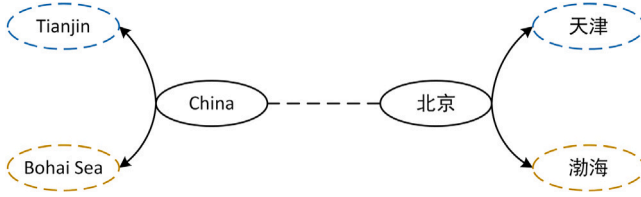


Fig. 2. An incorrect entity alignment.

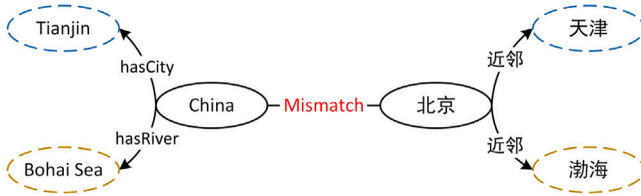


Fig. 3. Consider entity alignment of relations.

problems. The basic idea of this method is to model both entities and relations simultaneously by introducing a heterogeneous graph neural network. An iterative fusion method is also utilized to obtain a better entity representation. A concise version of this work has been successfully published in the IJCNN conference. See [16] for details. This paper increases the generalizability of the approach compared to the work at that time. Since not all datasets contain information about relations, this paper does not directly introduce a feature representation of relations. Instead, a relation-aware strategy is used to obtain the relations information, and the performance is slightly improved from the experimental results. Meanwhile, this paper also verifies the method's effectiveness from different angles through many experiments.

The contributions of this paper can be summarized as follows:

- (i) A relation-aware strategy is introduced to perceive the characteristic representation of a relation through entity information.
- (ii) An iterative fusion method is proposed by which the interaction between entities and relations can be better enhanced.
- (iii) The performance of the present model is shown to outperform the baseline model through extensive experiments on five public datasets.

2. Related work

Early approaches to entity alignment relied heavily on manual annotation with crowdsourcing techniques [17,18]. With the continuous development of the knowledge graph scale, manual annotation will consume a lot of time and human resources. Most existing entity alignment methods are based on embedding, and TransE [10] is one of the main techniques for entity embedding. MTransE proposed by Chen et al. [19] first embeds each knowledge graph into a separate low-dimensional vector space and then constructs transitions between different vector spaces. TransE can only model single-hop relations between entities and cannot handle complex multi-hop relations, thus losing important information. Therefore, Zhu et al. [20] proposed PTransE to embed entities and relations from different knowledge graphs into the joint semantic space based on the seed set. Instead of TransE, PTransE [21] is used to model multi-step relations. At the same time, high-confidence predicted aligned entities are filtered and added to the seed set to guide the subsequent entity alignment training. A truncated uniform negative sampling approach is designed in BootEA, proposed by Sun et al. [22]. From the original random selection of entities from all entities to replace the head or tail entity, it is changed to selecting one of several entities with the closest cosine distance to the head or tail entity. In addition, BootEA uses an iterative alignment method similar to IPTransE. However, BootEA can edit new entity pairs to reduce error propagation.

Graph Convolutional Networks (GCN) [23–25] is another major technique for KG embedding that has made significant progress in recent years. Wang et al. [26] proposed GCN-Align, which for the first time, uses a GCN model for the entity alignment task. The structural and entity attributes of the knowledge graph are embedded by two GCN models stacked in two layers, respectively. However, GCN can aggregate the neighbors' information around an entity well but ignores the relation information. RDGCN, proposed by Wu et al. [27] is a bi-graph convolutional structure constructing a dyadic graph for each knowledge graph. By alternating convolution operations on the original graph and the dyadic graph, the relation information in the knowledge graph is fully utilized to enhance the entity representation. The graph attention mechanism is applied in RDGCN, and the MuGNN proposed by Cao et al. [28] also applies the graph attention network. The knowledge graph is first complemented using rule knowledge, reducing heterogeneity between the two knowledge graphs. Then, two adjacency matrices are generated based on the self-attention and cross-graph attention channels and input into the GNN for embedding entities. Finally, a pooling function aggregates the two representations for entity alignment. Wu et al. [29] proposed a cross-graph neighborhood matching network NMN, which samples the entity neighbors to obtain the most informative neighbors. Their attention weights are computed based on the neighbor representation. Then, cross-graph matching vectors containing information about the entity and neighbor-matching information are generated for entity alignment. Most GNN-based approaches learn the embeddings of different KGs separately and ignore the useful pre-aligned links between two KGs. Xie et al. [30] proposed a new context-aligned augmented cross-graph attentional network CAEC-GAT for cross-lingual entity alignment tasks. This network can co-learn embeddings in different KGs by propagating cross-KG information through pre-aligned seed alignment. Also, Xie et al. [31] proposed DuGa-DIT, a dual-gated graph attention network with dynamic iterative training. Neighborhood and cross-KG alignment features are captured using intra-KG attention layers and cross-KG attention layers, and more cross-KG information is captured through a dynamic iterative process.

Recently, the emergence of several new approaches has led to significant progress in entity alignment. Wang et al. [32] proposed PEAMA, which utilizes multi-information aggregated persona entities for entity alignment. Huang et al. [33] proposed a new framework, RpAlign, which enables entity alignment between different KGs by predicting the relation of the entities. Lu et al. [34] presented a multi-neighborhood

sampling matching network MSM to address the challenge of structural heterogeneity through a new KG comparison network. Zhang et al. [35] proposed a neighboring entity filtering rule based on entity names and attributes for cross-language entity alignment. Using bi-relational graphs as supporting evidence in case of insufficient attribute information effectively promotes the close connection between entities. Li et al. [36] proposed a generative adversarial entity alignment method with stronger robustness to noisy data. An unsupervised entity alignment algorithm can automatically align entities without training labels. Chen et al. [37] proposed a locally inflated higher-order graph neural network HOLI-GNN for entity alignment. The effect of excessive smoothing due to neighborhood aggregation is mitigated by introducing a local expansion mechanism to amplify each feature of an entity. Li et al. [38] proposed a new joint unsupervised relation alignment model URAEA based on GCN for entity alignment. The model first adopts a novel approach to computing relation embeddings using entity embeddings, and then constructs unsupervised seeded relation alignments from these relation embeddings. Finally, entity alignment and relation alignment are performed together. Yang et al. [39] designed a novel entity-aligned relation mapping RMHN based on higher-order graph convolutional networks. By designing a novel higher-order GCN to aggregate higher-order information to obtain entity embeddings in bulk.

Most of the existing methods for the entity alignment task are based on homogeneous graphs for research. However, the actual knowledge graph is heterogeneous, and current approaches lack the study of heterogeneous graph neural networks. Therefore, we hope that a better representation of entity alignment tasks can be obtained by applying heterogeneous graph neural networks.

3. Methodology

In this section, the general framework of the entity alignment method based on relation-aware heterogeneous graph neural networks is presented in detail, and the model framework is shown in Fig. 4. It consists of three main parts:

- **Node Encoding Process.** Given entity names and descriptions, the entity names or descriptions are encoded as vectors by the pre-trained model BERT.¹ The feature representation of the relation is also perceived through the entity information using the relation-aware strategy.
- **Heterogeneous Graph Iterative Process.** An iterative fusion method is proposed to enhance the interaction of entity and relation nodes in heterogeneous graph neural networks.
- **Entity Alignment Process.** After obtaining the representation of entity nodes and relation nodes, a specific entity alignment step is performed by a distance function.

3.1. Node encoding process

In this paper, two types of semantic nodes are constructed for the proposed heterogeneous graph: entity nodes and relation nodes.

3.1.1. Entity nodes

For entity node e , this paper encodes the entity name/description using the BERT pre-training model. The CLS embedding of BERT is then filtered by a multilayer perceptron to obtain the initial entity representation $C(e)$:

$$C(e) = \text{MLP}(\text{CLS}(e)) \quad (1)$$

where CLS is the overall entity semantic representation vector of the BERT output, and MLP is the multilayer perception.

3.1.2. Relation nodes

To improve the generality of the method, this paper does not directly introduce the feature representation of the relation. Instead, it is obtained using the feature transformation of the relation's head entity and tail entity. Specifically, the relation embedding with weights is computed by applying a linear transformation to the entity embeddings and then calculating the relation embedding with weights. For relation r_k , the feature representation of the head entity r_k^h is computed as follows:

$$\alpha_{ij}^k = \frac{\exp(\text{LeakReLU}(\mathbf{a}^T [\mathbf{x}_i \mathbf{W}^h || \mathbf{x}_j \mathbf{W}^t]))}{\sum_{e_i \in \mathcal{H}_{r_k}} \sum_{e_j \in \mathcal{T}_{e_i r_k}} \exp(\text{LeakReLU}(\mathbf{a}^T [\mathbf{x}_i \mathbf{W}^h || \mathbf{x}_j \mathbf{W}^t]))} \quad (2)$$

$$\mathbf{r}_k^h = \text{ReLU} \left(\sum_{e_i \in \mathcal{H}_{r_k}} \sum_{e_j \in \mathcal{T}_{e_i r_k}} \alpha_{ij}^k \mathbf{x}_i \mathbf{W}^h \right) \quad (3)$$

α_{ij}^k represents the weight representation of the head entity e_i of relation r_k in the triplet, and $\mathcal{H}_{r_k}$ is the set of head entities of relation r_k in the triplet. $\mathcal{T}_{e_i r_k}$ is the set of tail entities with head entity e_i and relation r_k . $\mathbf{a}$ is a one-dimensional vector mapping a two-dimensional input to a scalar. $\mathbf{W}^h$ and $\mathbf{W}^t$ are the linear transfer matrices for the relation's head entity representation and tail entity representation, respectively.

By a similar process, one can compute the representation $\mathbf{r}_k^t$ of the tail entities and then add them to obtain the relation representation $\mathbf{r}_k$:

$$\mathbf{r}_k^t = \text{ReLU} \left(\sum_{e_j \in \mathcal{T}_{r_k}} \sum_{e_i \in \mathcal{H}_{r_k e_j}} \alpha_{ji}^k \mathbf{x}_j \mathbf{W}^t \right) \quad (4)$$

$$\mathbf{r}_k = \mathbf{r}_k^h + \mathbf{r}_k^t \quad (5)$$

3.2. Heterogeneous graph iterative process

In this section, the construction process and iterative fusion process of heterogeneous graphs are described in detail.

3.2.1. Initial construction of the heterogeneous graph

Two types of semantic node representations are given: entity nodes $E = \{e_i\}_{i=1}^N$ and relation nodes $R = \{r_j\}_{j=1}^M$, where N represents the number of entities, and M represents the number of relations. The initial construction of the heterogeneous graph G is shown below:

$$G = (E, R) \quad (6)$$

3.2.2. Nodes updating with iterative fusion

When the initial construction of the heterogeneous graph is completed, the relation nodes and entity nodes are updated and iteratively fused. As shown in Fig. 5, the relation nodes aggregate all the entity information directly adjacent to them. The updated relation node representation is used to update the entity nodes. Thus, iterative updates make the node representation more suitable for the entity alignment task.

Graph Attention Network (GAT) is used to complete the updating of nodes for better semantic fusion between entity nodes and relation nodes. In this paper, we give the updating process of relation and entity nodes in the l th layer of the iterative process. The updating process of relation nodes in the l th layer can be represented as:

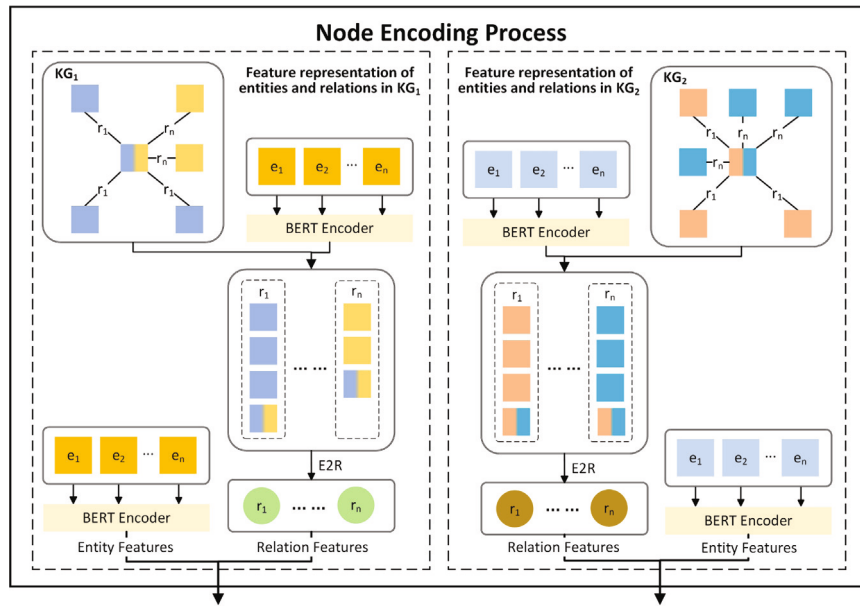
$$\tilde{\mathbf{r}}_j^{l+1} = \text{GAT} \left(\mathbf{r}_j^l, \{\mathbf{e}_i^l\}_{i \in \mathcal{N}_j} \right) \quad (7)$$

$$\mathbf{r}_j^{l+1} = \tilde{\mathbf{r}}_j^{l+1} \oplus \mathbf{r}_j^l \quad (8)$$

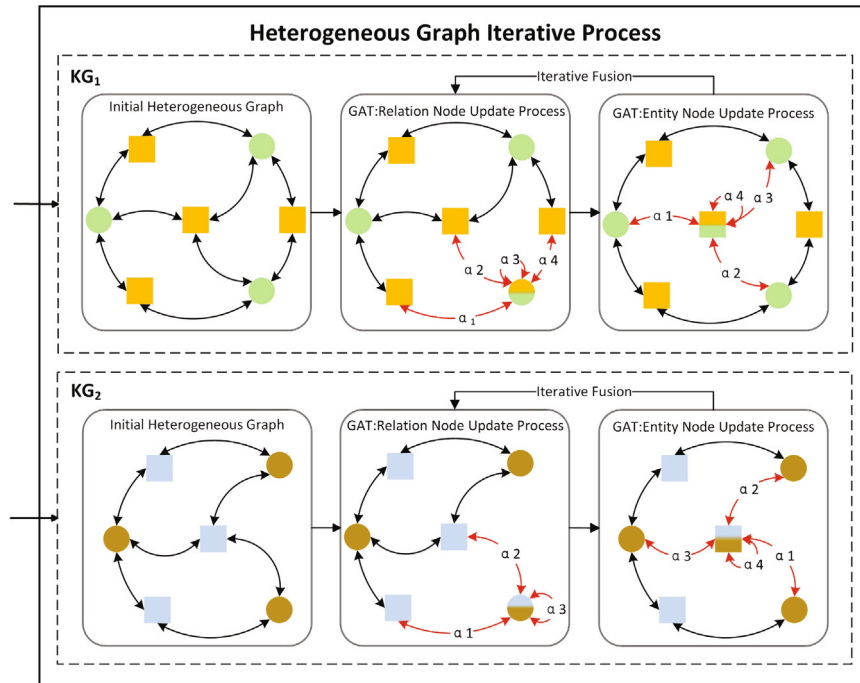
When the relation node update is completed, the updated relation node is used to update the entity node. The process of entity node update in layer l can be represented as:

$$\tilde{\mathbf{e}}_i^{l+1} = \text{GAT} \left(\mathbf{e}_i^l, \{\mathbf{r}_j^{l+1}\}_{j \in \mathcal{N}_i} \right) \quad (9)$$

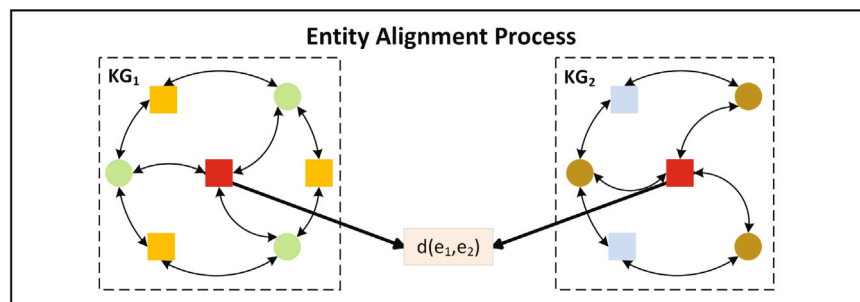
¹ <https://github.com/google-research/bert>



(a) Node Encoding Process



(b) Heterogeneous Graph Iterative Process



(c) Entity Alignment Process

Fig. 4. Overview framework of the model. The circles in the heterogeneity diagram represent relation-type nodes, and the squares represent entity-type nodes. The entity type nodes in red in the entity alignment process are the potentially matching entity pairs we want to find.

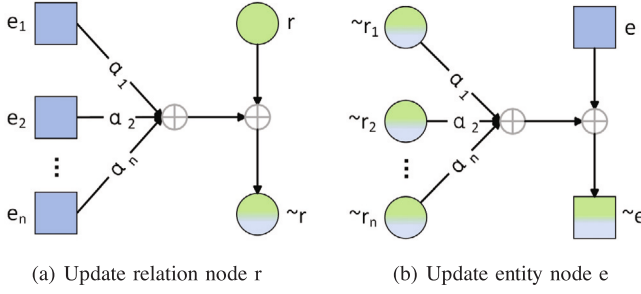


Fig. 5. Example of a one-round iterative fusion process.

$$\mathbf{e}_i^{l+1} = \tilde{\mathbf{e}}_i^{l+1} \oplus \mathbf{e}_i^l \quad (10)$$

In the updating process, this paper uses a graph attention network to update the representation of nodes. Take the entity node update as an example. Its specific update process is shown as follows:

$$c_{ij} = W_c [W_q \mathbf{e}_i \oplus W_k \mathbf{r}_j] \quad (11)$$

$$\alpha_{ij} = \frac{\exp(c_{ij})}{\sum_{l \in \mathcal{N}_i} \exp(c_{il})} \quad (12)$$

$$\tilde{\mathbf{e}}_i = \mathbf{e}_i + \sum_{j \in \mathcal{N}_i} \alpha_{ij} W_r \mathbf{r}_j \quad (13)$$

where $\oplus$ denotes the splicing between two vectors. W_c , W_q , W_k and W_r are trainable weight parameters. α_{ij} is the attention weight coefficient between $\mathbf{e}_i \in \mathbb{R}^{d_h}$ and $\mathbf{r}_j \in \mathbb{R}^{d_h}$. $\tilde{\mathbf{e}}_i$ is the updated entity node representation.

3.3. Entity alignment process

The distance between two entities is measured as a criterion for entity alignment, using the final entity representation X collected from the output of the heterogeneous graph neural network. Specifically, the distance $d(e_1, e_2)$ between two entities $e_1 \in G_1$ and $e_2 \in G_2$ can be calculated as:

$$d(e_1, e_2) = \|x_{e_1} - x_{e_2}\|_2 \quad (14)$$

where $\|x_{e_1} - x_{e_2}\|_2$ is the L_2 distance between the aligned entity pairs (e_1, e_2) .

3.4. Training

In this section, the loss function of the model and the overall training process will be defined. The detailed encoding, iteration, and training process are shown in Algorithm 1. The model's loss function consists of two components, the node encoding loss $loss_{node}$ and the entity alignment loss $loss_{align}$, each of which is described in detail below.

3.4.1. Node encoding loss

With the help of TransE [10], given a relation triple (e_h, r, e_t) , one wants to make $e_h + r \approx e_t$. Thus a scoring function $f(e_h, r, e_t) = \|e_h + r - e_t\|_2$ is defined to measure the plausibility of (e_h, r, e_t) , where $\|\cdot\|_2$ denotes the 2-parametric number. After TransE, an edge-based ranking loss function is used as the training target of the model, defined as:

$$loss_{node} = \sum_{(e_h, r, e_t) \in T} \sum_{(e'_h, r', e'_t) \in T'} [f(e_h, r, e_t) + \gamma_1 - f(e'_h, r', e'_t)]_+ \quad (15)$$

where $[\cdot]_+ = \max\{0, \cdot\}$ denotes the maximum value between 0 and the input. Entities e_h and e_t are the final entity representations after iterative fusion. Relation r is the final relation representation after

Algorithm 1: Encoding, iteration and training processes for heterogeneous graphs

Input: Source Knowledge Graph KG_1 and Target Knowledge Graph KG_2

Output: Entity embedding H'_e and relation embedding H'_r

```

1 Initialization:
2 for  $KG = (E_h, R, E_t)$  in  $KG_1, KG_2$  do
3    $H_e \leftarrow MLP(BERT(E_h, E_t))$  // Entity node encoding
4    $H_r \leftarrow Embedding(R)$  // Relation node encoding
5 end
6 Construction of heterogeneous graphs  $G_1$  and  $G_2$  for  $KG_1$  and  $KG_2$  respectively
7 Training:
8 for  $epo$  in  $range(epoch)$  do
9   Iterative fusion of entity and relation:
10  Compute  $H'_r \leftarrow f(H_e, H_r)$  // Update relation nodes
11   $H'_r \leftarrow H_r \cup H'_r$ 
12  Compute  $H'_e \leftarrow f(H_e, H'_r)$  // Update entity nodes
13   $H'_e \leftarrow H_e \cup H'_e$ 
14  Calculate the loss:
15  Evaluate the loss function  $loss_N$  by Eq. (15)
    // Calculating node encoding loss
16  Evaluate the loss function  $loss_A$  by Eq. (16)
    // Calculating entity alignment loss
17   $L \leftarrow loss_N + loss_A$ 
18  L.backward() // Perform backpropagation
19 end
20 return  $H'_e$  and  $H'_r$ 

```

iterative fusion. T' represents the set of negative sample triples of T . $\gamma_1 > 0$ is the edge hyperparameter separating positive and negative triples. T' is generated by destroying T . For example, for a triplet $(e_h, r, e_t) \in T$, a negative triplet (e'_h, r, e_t) is generated by replacing e_h , where e'_h also has the same relation type r connected to e_t , and the same substitution process for e_t .

3.4.2. Entity alignment loss

For entity alignment loss, the training objective of the model is to minimize the representation distance of equivalent entities on all marker alignments, similar to [40]:

$$loss_{align} = \sum_{(e_i, e_j) \in S} \sum_{(e'_i, e'_j) \in S'} [\text{dist}(e_i, e_j) + \gamma_2 - \text{dist}(e'_i, e'_j)]_+ \quad (16)$$

where $\text{dist}(e_i, e_j) = \|e_i - e_j\|_2$ is the L_2 distance between aligned entity pairs (e_i, e_j) . e_i and e_j are the final entity representations after iterative fusion. S' denotes the set of negative pairs of S generated by nearest neighbor sampling. The $\gamma_2 > 0$ is the edge hyperparameter. Because of the excellent performance of L_2 distance in experiments, this paper uses L_2 distance as a metric to search for the nearest negative samples.

3.4.3. Joint loss

The loss function of this model consists of two parts: the node encoding loss $loss_{node}$ and the entity alignment loss $loss_{align}$. The overall loss function $loss$ is calculated as follows:

$$loss = loss_{node} + loss_{align} \quad (17)$$

It is worth noting that different weights can be set for the node encoding loss $loss_{node}$ and entity alignment loss $loss_{align}$ to achieve balanced optimization. In this paper, considering that these two components are equally important, we set the same weight for them to treat both components equally in our experiments.

Table 1
DBP15K and DWY100K statistical information.

Datasets			Entities	Relations	Triplets
DBP15K	DBP15K _{ZH-EN}	Chinese	66,469	2,830	153,929
		English	98,125	2,317	237,674
	DBP15K _{JA-EN}	Japanese	65,744	2,043	164,373
		English	95,680	2,096	233,319
	DBP15K _{FR-EN}	French	66,858	1,379	192,191
		English	105,889	2,209	278,590
DWY100K	DWY100K _{WD}	DBpedia	100,000	330	463,294
		Wikidata	100,000	220	448,774
	DWY100K _{YG}	DBpedia	100,000	302	428,952
		YAGO3	100,000	31	502,563

4. Experiment

This paper evaluates the method's effectiveness on three cross-linguistic datasets from DBP15K and two large-scale datasets from DWY100K. The models are also further compared and analyzed from different perspectives.

4.1. Dataset

The model was evaluated by using the DBP15K dataset constructed by Sun et al. [41]. The DBP15K contains three large cross-linguistic datasets constructed from DBpedia. Use DBP15K_{ZH-EN} for Chinese to English, DBP15K_{JA-EN} for Japanese to English, and DBP15K_{FR-EN} for French to English. Each dataset contains 15,000 entity alignment seed pairs.

The effectiveness of the model on large-scale data was tested by performing experiments on DWY100K created by Sun et al. [22]. DWY100K contains two large-scale datasets, DWY100K_{WD} and DWY100K_{YG}, which are extracted from DBpedia, Wikidata, and YAGO3. DWY100K_{WD} denotes DBpedia to Wikidata, and DWY100K_{YG} denotes DBpedia to YAGO3. Each dataset has 100,000 entity-aligned seed pairs. Table 1 lists the details of each dataset.

4.2. Experiment settings

For all baseline methods, 30% of entity alignment seed pairs were randomly used as training data and the rest as test data. Two evaluation metrics are used in this task: MRR and Hits@N. MRR denotes the average inverse rank of all correct entities, and Hits@N represents the proportion of right entities whose rank is not greater than N. Higher values of MRR and Hits@N indicate a better entity alignment model.

For the parameter settings, the input dimension of Bert is set to 768, and the output dimension is set to 300. The dimension of the GAT hidden unit is set to 300. The learning rate is set to 0.005. The experiment results shown in this paper are after 1000 rounds of iterative fusion of entities and relations.

4.3. Baselines and comparison result

Table 2 reports the experimental results of the different models on three large cross-linguistic datasets of DBP15K, along with the means and standard deviation results of the methods. In this paper, the baseline models are classified into three categories: models based on TransE, models based on GCN, and models based on others. The results show that the proposed method performs well on all three datasets, which indicates that the proposed method is effective.

This paper's present model performs better when compared with recent work. It is worth noting that although DuGa-DIT slightly outperforms the present model in some of the results on the three datasets, the present model significantly outperforms DuGa-DIT regarding operational efficiency. Specific experimental results are shown in Table 5.

Table 4 reports the experimental results of the model on the large-scale dataset DWY100K. The experimental results show that the present model performs better on the large-scale dataset. The present model produces better results for cross-linguistic and large-scale data, combining the experimental results in Tables 2 and 4.

To evaluate the efficiency of this model, this paper compares the prediction time with a good baseline model on the DBP15K_{FR-EN} dataset. From Table 5, our method produces better results in terms of efficiency and performance.

4.4. Sensitivity to previously aligned ratios

In this paper, we validate the method's effectiveness by comparing it with the baseline models RDGCN [27] and KECG [40] at different previously aligned ratios. The results show that the present method outperforms the compared baseline models in Fig. 6. The experimental results of the present method are slightly worse than those of RDGCN when less training data is used. This is because introducing two types of nodes into the present model requires a certain amount of training data to be iteratively fused to obtain better features. However, as the amount of data increases, the advantages of heterogeneous graphs gradually appear, and their growth rate is faster through the iterative fusion of entities and relations.

4.5. Ablation study

This paper conducts ablation studies on the model using three cross-linguistic datasets from DBP15K, validating the effectiveness of the main components in the model. The specific experimental results of the ablation studies are shown in Table 3.

In this paper, a total of three different combinations are performed to verify the effectiveness of BERT encoding and heterogeneous graphs. These include the combination of one-hot encoding and heterogeneous graph neural network (one-hot+HGNN), the combination of BERT encoding and graph convolutional neural network (BERT+GCN), and the combination of BERT encoding and graph attention network (BERT+GAT). The experimental results show that the essential components of this paper are effective.

4.6. Visualization of attention scores

In this paper, we observe the weight parameters of the edges in GAT by visualizing the attention mechanism used in the iterative process. Since the adjacency matrix generated in the construction of the graph is sparse, only a central entity and the entities directly connected to it are randomly selected for display. In this paper, we take the DBP15K_{ZH-EN} dataset as an example and select a set of Chinese-English entity alignment seed pairs to display, and the visualization results are shown in Fig. 7. Since the attention coefficients are relatively small, the visualization results are the results after the coefficients are magnified ten times.

The attention visualization is selected from the direct first-order neighbors of each central entity. As can be seen in Fig. 7, the attention mechanism is beneficial for aligning two central entities. The model produces higher weighting coefficients for the common neighbors of the central entities and lower weighting coefficients for their unique neighbors. For example, the “共和党(美国)” entity in Fig. 7(a) and the “Saint Lawrence River” entity in Fig. 7(b) have lower weight coefficients as their unique neighbors. This effectively alleviates the model's sensitivity to the structural differences of KG. This example clearly illustrates that the attention mechanism in this model can accurately find the more relevant features in the entity pairs.

Table 2

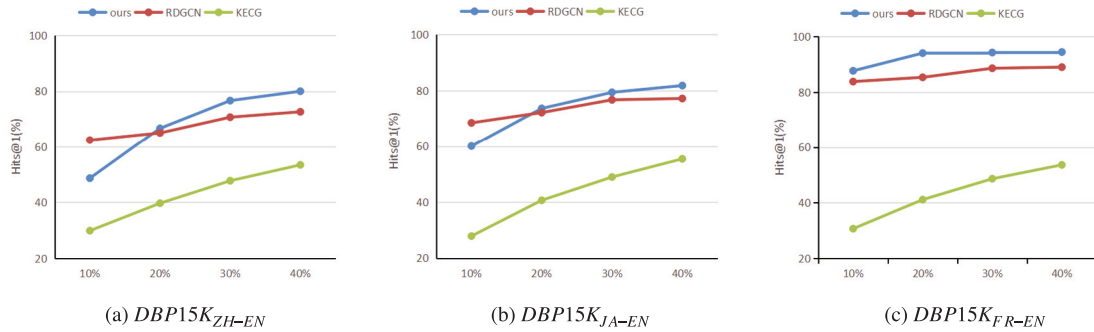
Baseline model comparison results in dataset *DBP15K*. The method means and standard deviations are reported in this paper. “-” indicates not reported in the original paper. Scores marked in **bold** indicate the best performance among all methods.

Method		<i>DBP15K</i>								
		<i>DBP15K_{JA-EN}</i>			<i>DBP15K_{ZH-EN}</i>			<i>DBP15K_{FR-EN}</i>		
		Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
TransE	MTransE [11]	0.279	0.575	0.349	0.308	0.614	0.364	0.244	0.556	0.335
	JAPE [41]	0.363	0.685	0.476	0.412	0.745	0.490	0.324	0.667	0.430
	BootEA [22]	0.622	0.854	0.701	0.629	0.848	0.703	0.653	0.874	0.731
GCN	KECG [40]	0.490	0.844	0.610	0.478	0.835	0.598	0.486	0.851	0.610
	AliNet [29]	0.549	0.831	0.645	0.539	0.826	0.628	0.552	0.852	0.657
	RDGCN [27]	0.767	0.895	0.812	0.708	0.846	0.746	0.886	0.957	0.911
	MRAGEA [42]	0.757	0.933	0.826	0.757	0.929	0.827	0.780	0.948	0.849
	HMAN [33]	0.557	0.860	0.674	0.562	0.851	0.670	0.550	0.876	0.666
	MHGCN [43]	0.767	0.891	0.821	0.732	0.867	0.793	0.864	0.923	0.898
	EVA [44]	0.762	0.913	0.817	0.761	0.907	0.814	0.793	0.942	0.847
	AttrGNN [45]	0.783	0.921	0.834	0.796	0.929	0.845	0.918	0.977	0.910
	SSP [46]	0.721	0.935	0.800	0.739	0.925	0.808	0.739	0.947	0.818
	DuGa-DIT [31]	0.914	0.952	0.928	0.807	0.882	0.832	0.982	0.992	0.985
	CAECGAT [30]	0.836	0.956	0.881	0.756	0.934	0.818	0.947	0.992	0.965
	DRG+ESGCN [35]	0.807	0.933	0.853	0.757	0.907	0.811	0.970	0.973	0.927
	GAEA [36]	0.846	0.925	0.846	0.774	0.875	0.813	0.914	0.957	0.902
Others	PEAMA [32]	0.551	-	0.634	0.562	-	0.644	0.558	-	0.645
	RpAlign [33]	0.730	0.890	0.782	0.748	0.889	0.794	0.752	0.900	0.801
	MSM [34]	0.788	0.919	-	0.708	0.867	-	0.906	0.973	-
ours		0.799 ±0.002	0.972 ±0.005	0.864 ±0.003	0.768 ±0.004	0.963 ±0.002	0.843 ±0.004	0.939 ±0.003	0.992 ±0.006	0.959 ±0.004

Table 3

Comparison results of ablation studies in dataset *DBP15K*.

Model	<i>DBP15K_{JA-EN}</i>			<i>DBP15K_{ZH-EN}</i>			<i>DBP15K_{FR-EN}</i>		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
ours(BERT+HGNN)	0.799	0.972	0.864	0.768	0.963	0.843	0.939	0.992	0.959
one-hot+HGNN	0.761	0.921	0.821	0.694	0.889	0.766	0.768	0.951	0.839
BERT+GCN	0.577	0.912	0.698	0.581	0.914	0.702	0.685	0.943	0.781
BERT+GAT	0.677	0.937	0.769	0.652	0.931	0.751	0.949	0.992	0.966

**Fig. 6.** Experimental results for different alignment ratios.

4.7. Visualization of entity features

The entity characteristics of the initial and final stages were visualized to verify the model's validity. To see the changes in entity feature distribution more intuitively, this paper uses the high-dimensional data reduction algorithm t-distributed random neighbor embedding (t-SNE) to map the feature distribution in two-dimensional space. The entity feature distribution by applying this method is shown in Fig. 8. Among them, Fig. 8(a), Fig. 8(c), and Fig. 8(e) show the initial entity feature visualization results for the datasets *DBP15K_{ZH-EN}*, *DBP15K_{JA-EN}*, and *DBP15K_{FR-EN}*, respectively. Fig. 8(b), Fig. 8(d), and Fig. 8(f) show the visualization results of entity features of datasets *DBP15K_{ZH-EN}*, *DBP15K_{JA-EN}* and *DBP15K_{FR-EN}* after 1000 rounds of entity and relation iterative fusion, respectively.

As can be seen in Fig. 8(a), Fig. 8(c), and Fig. 8(e), the overall distribution of initial entity features on the three datasets is irregular. As can be seen in Fig. 8(b), Fig. 8(d), and Fig. 8(f), when the entity and relation

are iteratively fused for 1000 rounds of training, the entity pairs appear significantly aligned on all three datasets. t-SNE visualization also indirectly proves the effectiveness of this model.

4.8. Case study

To demonstrate the effectiveness of this methodology, a case study is presented in this section for in-depth analysis. Fig. 9 shows the results of the case study compared to a strong baseline. The present method

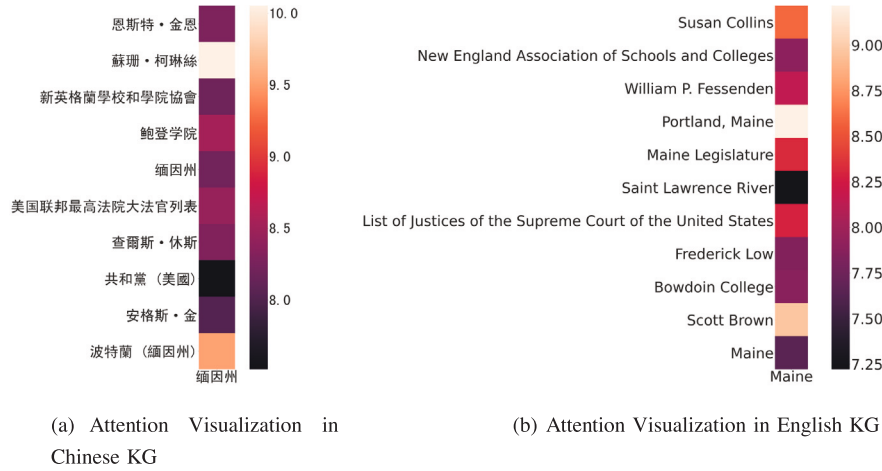


Fig. 7. The results of the attention visualization are plotted. The central entities “緬因州” and “Maine” are a set of entity alignment seed pairs in the $DBP15K_{ZH-EN}$ dataset.

Table 4

Baseline model comparison results in dataset $DWY100K$. This paper reports the means and standard deviations of the methods. Scores marked in **bold** indicate the best performance among all methods.

Method		$DWY100K$					
		$DWY100K_{WD}$			$DWY100K_{YG}$		
		Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
TransE	MTransE [11]	0.281	0.520	0.363	0.252	0.493	0.334
	JAPE [41]	0.318	0.589	0.411	0.236	0.484	0.320
	BootEA [22]	0.748	0.898	0.801	0.761	0.894	0.808
GCN	GCN-Align [26]	0.506 ^a	0.772 ^a	0.600 ^a	0.597 ^a	0.838 ^a	0.682 ^a
	KECG [40]	0.632	0.900	0.726	0.728	0.915	0.795
	AliNet [29]	0.690	0.908	0.766	0.786	0.943	0.841
	MRAEA [42]	0.794	0.930	0.856	0.819	0.951	0.875
	SSP [46]	0.772	0.960	0.842	0.811	0.968	0.869
RpAlign [33]		0.826	0.934	0.862	0.838	0.945	0.872
ours		0.832	0.958	0.873	0.854	0.977	0.894
		± 0.003	± 0.003	± 0.002	± 0.004	± 0.002	± 0.003

^a Indicates metrics not reported in the paper. The results shown in this paper are from Li et al. [40].

Table 5

Prediction times for different models on dataset $DBP15K_{FR-EN}$.

Methods	Time
GM-EHD-JEA	1,474 s
RDGCN	45 s
HGCN-JE	50 s
NMN	84 s
CAECGAT	42 s
DuGa-DIT	44 s
ours	8 s

The baseline modeling experimental results shown in this paper were provided by Xie et al [31].

achieves better results. We use the highest ranked entity as the alignment centre entity for model prediction, and the experimental results of 3 samples on different models are shown in Fig. 9. The present model obtained correct alignment results in these samples, again proving the importance of introducing relations. When the central entity has more neighbor nodes (as in cases 1 and 2), the relation information can help us distinguish different entities. When the central entity has fewer neighbor nodes (as in the 3rd case), the relation information can help us

to add helpful information. Therefore, the components of this method have an essential role to play.

5. Conclusion

This paper presents an entity alignment method based on relation-aware heterogeneous graph neural networks. The method uses heterogeneous graph neural networks to model entities and relations simultaneously. It uses an iterative fusion method to enhance the interaction between the two types of semantic nodes. Since not all datasets contain information about relations, this paper does not directly introduce information about relations. The generalizability of the approach is improved by using a relation-aware strategy to obtain relations from entity perception. In this paper, the method’s effectiveness is evaluated using the cross-lingual dataset $DBP15K$ and the large-scale dataset $DWY100K$. The results show that better performance is obtained by improving all metrics compared to the existing methods.

Attribute information is an essential class of information in knowledge graphs, and the advantages of this method are not apparent when the data relations are more complex. In our future work, we will consider introducing attribute information to further improve the method’s performance.

CRedit authorship contribution statement

Zirui Zhang: Writing – original draft, Methodology, Conceptualization. **Yiyu Yang:** Writing – review & editing. **Benhui Chen:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

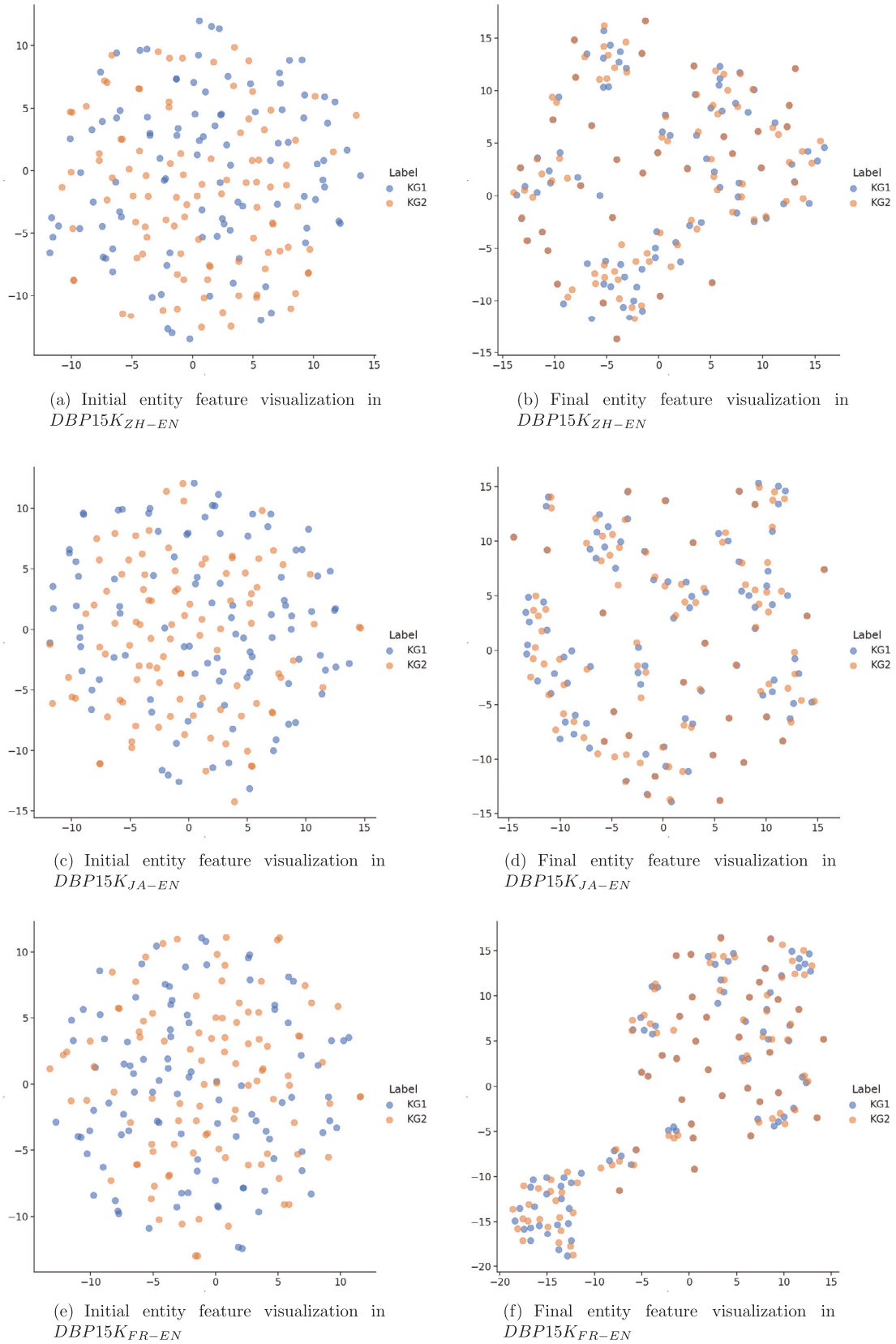


Fig. 8. Visual presentation of entity features in the initial and final stages.

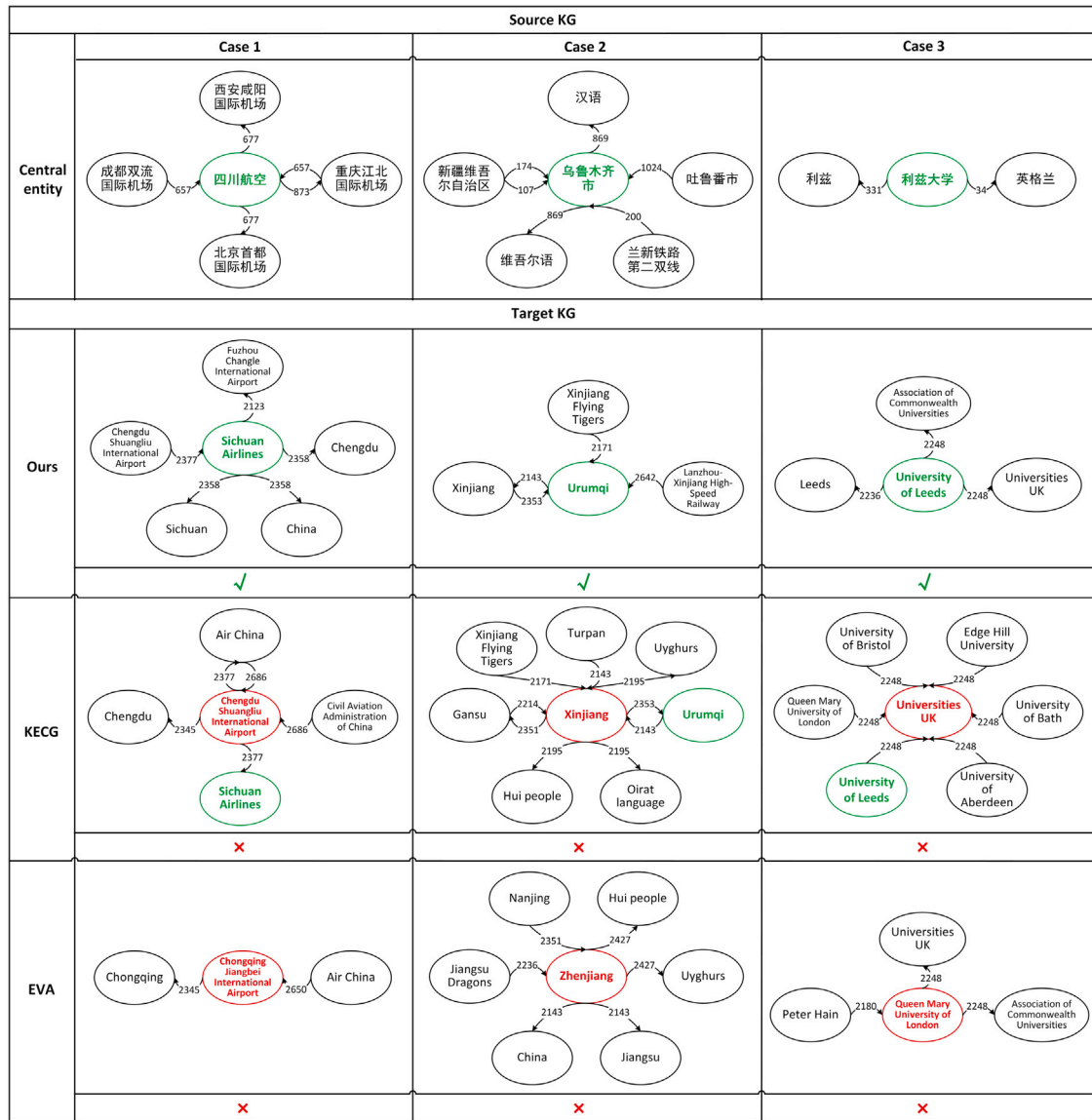


Fig. 9. Case study on the dataset $DBP15K_{ZH-EN}$. Green entities indicate correctly aligned central entities, and red entities indicate misaligned central entities.

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